Key Features of Exponential Functions

Name _____

Algebra 2 Lesson 6-1 (fast track) Date _____ Period _____

Learning goal: I can tell how a reflection or a shift moves the y-intercept and the asymptote of an exponential function, and compare two exponential functions by their growth or decay factors.

Language goal: I can explain an exponential function using the words *growth factor*, *decay factor*, *y-intercept*, *asymptote*, and *reflection*.

On the quiz: (1) Example 2: which function is 3^x after a vertical stretch by 8 and a reflection across the x-axis? (2) Growth or decay after 5 years (full Savvas quiz only). (3, bonus) Domain, range, y-intercept, asymptote of $\frac{1}{2} \cdot 6^x$ (Example 2's rules). (4) Write and graph a decay model (full Savvas quiz only). (5) Example 5: which function has a greater growth factor than a graph?

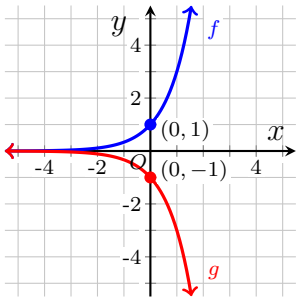
On the Topic 6 test: Question 1 asks which function is 5^x stretched by 2 and reflected across the x-axis (Example 2). Question 4 asks which function decays faster than a graphed curve (Example 5). Question 17 asks for the y-intercept and asymptote of $-4(0.3)^x + 1$ (Example 2; Form B's $2(3)^{-x}$ is Example 1, which this packet skips).

Words: exponential function (*función exponencial*), growth factor (*factor de incremento*), decay factor (*factor de decremento*). An **exponential function** is $f(x) = a \cdot b^x$ with $a \neq 0$, $b > 0$, $b \neq 1$; its y-intercept is a because $b^0 = 1$.

Example 2: Graph Transformations of Exponential Functions

Graph each function. Describe the graph in terms of transformations of the parent function $f(x) = 3^x$. How do the asymptote and y-intercept of the given function compare to the asymptote and y-intercept of the parent function?

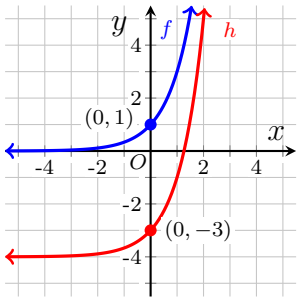
A. $g(x) = -3^x$



$g(x) = -3^x = -f(x)$: multiplying by -1 **reflects** the graph across the x-axis.

- y-intercept: moves from a to $-a$, so from 1 to -1
- Asymptote: does not change; still the x-axis ($y = 0$)

B. $h(x) = 3^x - 4$



$h(x) = 3^x - 4 = f(x) - 4$: adding a constant k **shifts** the graph vertically by k units (here down 4).

- y-intercept: moves from 1 to $1 + k$, so to -3
- Asymptote: also moves; it is $y = k$, or $y = -4$

Say it: For $h(x) = 3^x - 4$ the *y-intercept* moves from 1 to _____ and the *asymptote* moves from $y = 0$ to _____. For $g(x) = -3^x$ the *reflection* moves the *y-intercept* to _____, but the *asymptote* _____.

Try It 2: Compare the asymptote and y-intercept

How do the asymptote and y-intercept of the given function compare to the asymptote and y-intercept of the function $f(x) = 5^x$?

Hint: $f(x) = 5^x$ has y-intercept 1 and asymptote $y = 0$. Put $x = 0$ to find each new y-intercept.

a. $g(x) = 5^{x+3}$

y-intercept: $g(0) =$ _____ Asymptote: $y =$ _____

Compared with f:

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b. $h(x) = 5^{-x} + 6$

y-intercept: $h(0) =$ _____ Asymptote: $y =$ _____

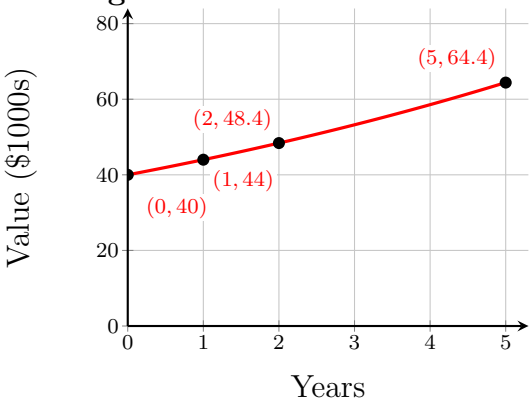
Compared with f (which transformations?):

Example 5: Compare Two Exponential Functions

A museum purchased a painting and a sculpture in the same year. Their values are changing exponentially as shown. Which art work's value is increasing more quickly?

Sculpture value: $f(x) = 50(1.075)^x$, in thousands of dollars in x years.

Painting value:



Identify key features for the sculpture: $f(x) = 50(1.075)^x$. The initial purchase price is \$50,000. The **growth factor** is 1.075.

Find the growth factor, b, for the painting. The ratio of y-values one year apart is b:

$$b = \frac{44}{40} = 1.1 \quad \text{and} \quad b = \frac{48.4}{44} = 1.1$$

$$g(x) = 40(1.1)^x$$

Verify for $x = 5$: $g(5) = 40(1.1)^5 \approx 40(1.6105) \approx 64.4$.

The growth factor for the painting is 1.10. Since $1.10 > 1.075$, the painting's value is increasing more rapidly.

Say it: The painting's *growth factor* is _____ because $44 \div 40 =$ _____; the sculpture's is _____. The painting's value grows faster because _____. A *decay factor* of 0.6 makes a value shrink faster than a *decay factor* of 0.75 because ...

Try It 5: When does the painting pass the sculpture?

In Example 5, in what year will the value of the painting become greater than the value of the sculpture?

Hint: compare $f(x) = 50(1.075)^x$ and $g(x) = 40(1.1)^x$ year by year (thousands of dollars, to the nearest tenth).

x (years)	Sculpture $f(x)$	Painting $g(x)$	Which is greater?
0	50	40	sculpture
5			
8			
9			
10			

The value of the painting becomes greater in year _____, because

IXL: Practice for the quiz

SmartScore 70 or above on each starred (★) skill replaces your quiz score.

Code	Skill	Feeds
LWE	Evaluate exponential functions	warm-up (Try It 5 table)
RSN ★	Transformations of functions	Quiz 1
EFD ★	Domain and range of exponential functions	Quiz 3 (bonus)
2YT ★	Exponential functions over unit intervals	Quiz 5

Exit question: Model with mathematics

A colony of bacteria starts with 50 organisms and quadruples each day. Write a function, $P(t)$, that represents the population of the bacteria after t days. Then find the number of bacteria that will be in the colony after 5 days.

Starting amount (y-intercept): _____ *Growth factor*: _____ because _____

$P(t) =$ _____ After 5 days: _____ bacteria.

Is this exponential growth or decay? Compare its *growth factor* with the painting's *growth factor* in Example 5.